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SECTION 1: VIBRATION TRIPPING CRITERIA & RUNNING CLEARANCES

1.1 Radial Shaft Vibration Thresholds (Proximity Probes):

- **Alert Alarm: 4.5 mm/s RMS (or 45 micrometers peak-to-peak).**
- Emergency Automatic Trip: 7.1 mm/s RMS (or 65 micrometers peak-to-peak).
- Spectral Analysis Protocol: Any presence of sub-synchronous peaks between 0.40X and 0.48X running speed indicates oil whirl; immediately verify lube temp.

SECTION 2: LUBRICATING OIL & BEARING TEMPERATURE SPECIFICATIONS

2.1 Lubricant Spec: ISO VG 46 Turbine Oil with rust/oxidation inhibitors (ASTM D4304).

2.2 Lube Oil Header Pressure: 1.8 to 2.4 bar gauge. Filter Delta-P limit: 1.0 bar max.

2.3 Bearing Temperature Limits:

- **Journal Bearings: Max allowable 90 deg C. Trip at 100 deg C.**
- Active Thrust Bearing Face: Max continuous 88 deg C. Trip at 98 deg C.

SECTION 3: DRY GAS SEALS (DGS) & BARRIER SYSTEM

3.1 Buffer Gas (N2) Supply Pressure: Must be minimum 3.0 bar above internal seal cavity.

3.2 Primary vent leakage exceeding 12 Nm3/h indicates primary seal face micro-scoring.

SECTION 4: TROUBLESHOOTING GUIDE (FMEA CROSS-REFERENCE)

Failure Mode: High Radial Vibration + Oil Whirl.

Corrective Actions: (a) Increase lube oil supply temp to 45C to reduce viscosity,

(b) Verify tilting pad pivot clearances, (c) Check shaft balance & alignment.